Data Calibration of the Actual Versus the Theoretical Micro Electro Mechanical Systems (MEMS) Based Accelerometer Reading Through Remote Monitoring of Padre Jacinto Zamora Flyover

John Mark G. Payawal
School of Civil, Environmental, and Geological Engineering
Mapua Institute of Technology
Intramuros, Manila, Philippines

Francis Aldrine A. Uy
School of Civil, Environmental, and Geological Engineering
Mapua Institute of Technology
Intramuros, Manila, Philippines

John Paul D. Carreon
School of Civil, Environmental, and Geological Engineering
Mapua Institute of Technology
Intramuros, Manila, Philippines

Abstract— **This paper shows the application of Structural Health Monitoring, SHM into bridges. Bridges are structures built to provide passage over a physical obstruction such as rivers, chasms or roads. The Philippines has a total of 8,166 national bridges as published on the 2015 atlas of the Department of Public Works and Highways (DPWH) and only 2,924 or 35.81% of these bridges are in good condition. As a result, PHP 30.464 billion of the 2016 budget of DPWH is allocated on roads and/or bridges maintenance alone. Intensive spending is owed to the present practice of outdated manual inspection and assessment, and poor structural health monitoring of Philippine infrastructures. As the School of Civil, Environmental, & Geological Engineering of Mapua Institute of Technology (MIT) continuous its well driven passion in research based projects, a partnership with the Department of Science and Technology (DOST) and the DPWH launched the application of Structural Health Monitoring, (SHM) in Padre Jacinto Zamora Flyover. The flyover is located along Nagtahan Boulevard in Sta. Mesa, Manila that connects Brgy. 411 and Brgy. 635. It gives service to vehicles going from Lacson Avenue to Mabini Bridge passing over Legarda Flyover. The flyover is chosen among the many located bridges in Metro Manila as the focus of the pilot testing due to its site accessibility, and complete structural built plans and specifications necessary for SHM as provided by the Bureau of Design, BOD department of DPWH. This paper focuses on providing a method to calibrate theoretical readings from STAAD Vi8 Pro and sync the data to actual MEMS accelerometer readings. It is observed that while the design standards used in constructing the flyover was reflected on the model, actual readings of MEMS accelerometer display a large difference compared to the theoretical data ran and taken from STAAD Vi8 Pro. In achieving true seismic response of the modeled bridge or hence syncing the theoretical data to the actual sensor reading also called as the independent variable of this paper, analysis using single degree of freedom (SDOF) of the flyover under free vibration without damping using STAAD Vi8 Pro is done. The earthquake excitation and bridge responses is subjected to earthquake ground motion in the form of ground acceleration or Peak Ground Acceleration, PGA. Translational acceleration load is used to simulate the ground motion of the time history analysis acceleration record in STAAD Vi8 Pro. Though sensors measure three axes simultatniously, SDAF will be used in the analysis since this paper focuses in the up and down movement of the bridge being inspected. Lastly, environmental factors such as wind and rainfall were all viewed as external elements. With this, upon calibration, these factors were corrected not to affect the readings of the sensors used in the study.**

Keywords— accelerometer, analysis using single degree of freedom, micro electro mechanical system, peak ground acceleration, structural health monitoring.

I. INTRODUCTION

Structural Health Monitoring (SHM) is a process of identifying damage present in constituent materials of different engineering infrastructures such as civil, aeronautical or mechanical engineering. This process aims to ensure that the structure must always remain still, within the specified design criteria under a certain period. Most SHM in Civil

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Engineering is done on vertical infrastructure, specifically on high-rise commercial buildings. They are prone to damage from normal aging due to continuous usage, changes in the environment and by accidental events. This paper shows the application of SHM into bridges. Bridges are structures built to provide passage over a physical obstruction such as rivers, chasms or roads. Application of SHM to bridges requires usage of smart sensors that enables the user to identify potential damage caused by vehicles traveling across the bridge.

SHM of Jindo Bridge in Japan is a large scale of wireless smart sensor network (WSSN) installed to continuously monitor various physical states of the structure. It served as a test bed for evaluating wireless sensor network technology for SHM and inspiration of this paper. The monitoring system installed can independently estimate different physical states of the bridge such as modal properties like natural frequencies, modal damping and mode shapes. This can be obtained from measured accelerations and which is numerically modeled to determine the performance of structures and locate possible areas for damage. For Jindo Bridge, wind velocity fluctuations were measured through ultrasonic anemometers installed. The data obtained was processed and enabled spatial and temporal variation which were analyzed and interpreted to show an average in the aerodynamic admittance effect that recommended a significant enhancement in evaluating aerodynamic safety and serviceability of the bridge.

The allocated budget for road transportation of the Philippines is P195.1 billion in the year 2015 (The President's Budget Message Fiscal Year 2015, 2014). In the present day, most of these infrastructures are about to reach their design lives and will need retrofitting to remain in service. Bridge structures are the most common type of structure that experiences problems regarding its design life. There are 8,166 existing national bridges in the Philippines according to the December 2015 report of the Department of Public Works and Highways (DPWH, 2015). Some of these bridges are already considered structurally deficient and need to be replaced or retrofitted to prevent putting the public at risk. Out of 8,166 bridges in the Philippines, only 2,924 of which are rated to be in "good" condition and that is about 35.81% of the total. About 3,608 (44.18%) are in "fair" condition, 1,070 (13.10%) are in "poor" condition, 438 (5.36%) are in "bad" condition and 126 (1.54%) are still on further assessment as stated in the summary of bridge condition for calendar year 2015 of DPWH (DWPH, 2015).

As the School of Civil, Environmental, & Geological Engineering of Mapua Institute of Technology (MIT) continuous its well driven passion in research based projects, a partnership with the Department of Science and Technology (DOST) and the DPWH launch the application of SHM in Padre Jacinto Zamora Flyover. The flyover is located along Nagtahan Boulevard in Sta. Mesa, Manila that connects Brgy. 411 and Brgy. 635. It gives service to vehicles going from Lacson Avenue to Mabini Bridge passing over Legarda Flyover. Padre Jacinto Zamora Flyover is chosen among the many bridges located in Manila as the focus of the pilot testing due to its accessibility, and complete structural built plans and specifications necessary for SHM.

A. Research Aim/s

The research primarily aims to provide a method to calibrate theoretical readings from STAAD Vi8 Pro and sync the data to actual MEMS accelerometer readings. It is observed that while the design standards used in constructing the flyover was reflected on the model, actual reading of MEMS accelerometer possess a large difference compared to theoretical data from STAAD Vi8 Pro. To do this, secondary objectives are established and are as follows:

1. To create a step-by-step procedure on how sensor calibration was done and how the flyover was modeled in STAAD Vi8 Pro,
   - A detailed documentation of the modeling will be provided which is necessary in founding a credible design method for SHM of bridges.
2. To provide a summary of all the DPWH standards used in the construction of the Flyover and,
   - The list will extend from materials used up to the design criteria at the time the flyover was built.
   - These summarizes the construction of sub-structure and the super-structure of Padre Jacinto Zamora Bridge,
3. To show fine-tuning and calibration techniques and methods done on MEMS accelerometers.

Actual reading of MEMS accelerometer as they are installed in the flyover is treated as true value and treated as independent parameters of the study. This section is written in Chapter II: Review of Related Literature.

B. Research Limits

The effective method in calibrating the theoretical MEMS accelerometers to actual reading is limited to the following:

1. The source of data for SHM is the Padre Jacinto Zamora Flyover - a concrete flyover,
2. The development of MEMS accelerometers will not be discussed into full details but will be included in the literature review (*the development of the sensors is done by the School of EECE of MIT*) and,
3. The method of calibration is assumed to apply only when modeling is done under STAAD Vi8 Pro.

II. STRUCTURAL HEALH MONITORING

Structural health monitoring (SHM) is usually viewed as a simple data acquisition from a structure in response to its external and internal excitations. More importantly, SHM includes the analysis of this data to measure the change in state of the structure and from that formulate a diagnosis of important information such as capacity of the structure and its remaining service life (Dong et al, 2010). The aim of structural health monitoring is to acquire knowledge of the capacity of operational structures on a continuous real-time basis according to Chang (1999). Maintenance and inspections at fixed intervals provide limited information about the structural condition of the system under monitoring. These methods are usually more expensive and require extensive labor. However, due to the advances in the modern technology enable the

development of real-time inspection and damage monitoring. Hence, the importance of structural health monitoring is the application of independent systems for the constant monitoring, inspection and damage detection of structures involving minimum labor. (Gastineau et al, 2009).

Chang et al (2003) focus the aim of structural health monitoring to locating and quantifying the severity of the damage in structures such as buildings and bridges as they happen. Chang et al also adopt the term "global health monitoring" as methods of determining the existence of damage within a structure without locating it. Furthermore, Chang et al refer to non-destructive methods as "local health monitoring" methods due to the fact that it can indicate the location of the damage once the presence of the damage has been established by the "global" methods.

According to Worden and Dulieu-Barton (2004), structural health monitoring refers to monitoring the behavior of structures through the use of sensing networks while they are in service. On the other hand, Worden and Dulieu-Barton defines non-destructive evaluation as the characterization and indicating the location of damage after temporarily taking the structure out of service. Worden and Dulieu-Barton further discussed the definition of structural health monitoring by establishing five principal problems with the identification of damage: (1) damage detection, (2) identifying its possible location, (3) classifying the damage type, (4) damage extent assessment and (5) quantifying the remaining service life of the structure.

As technology develops, the objectives and definition of structural health monitoring also evolve and change. Various definitions for structural health monitoring have been proposed and they go along with several ideas that are essential for structural health monitoring. These ideas are (1) real-time monitoring of (2) currently in-service structures utilizing (3) a system of sensors to gather data that can be used to (4) represent variations in the condition of a structure over a period of time. The collected data is then (5) transmitted over a network, and (6) data processing algorithms can be used for locating, categorization and evaluation of the damage, as well as predicting the remaining service life of the structure (Gastineau et al, 2009).

Structural Health Monitoring is applied to many different engineering fields such as Aeronautical, civil and mechanical engineering. The Aeronautical Engineers began to apply vibration-based damage identification between the 1970's and the 1980's in conjunction to the development of the space shuttle. The Shuttle Modal Inspection System or SMIS was developed to locate fatigue damage in different components such as control surfaces, fuselage panels and lifting surface. It is mandatory to always apply health monitoring in these areas since they are covered with thermal protection system making them inaccessible which makes it inapplicable to conventional local non-destructive examination methods. Application of SMIS has been successful in locating damaged components since 1987 and many researchers have risen to meet other challenges faced with orbiter shuttles. (Farrar & Worden, 2006)

Most common application of Structural Health Monitoring can be found in the fields of civil engineering. Civil engineers used vibration-based assessment of both buildings and bridges since the 1980's to determine modal properties and quantities such as modal shape curvature and dynamic flexibility matrix indices that are one of the main features used to identify damages present in the bridge structure.

Bridge structures are exposed to numerous factors that can affect its health such as degradation due to aging, environmental hindrances and excess loading. Bridge Structural Health Monitoring is a maintenance strategy used to ensure that the design capacity of the bridge structure is continuously met to prevent damage that can cause failure and affect safety and efficiency of the bridge. (Zhu & Law, 2015). Moreover, it has been a subject of research for more than 30 years and until now is continuously developed and applied to different bridges all over the globe.

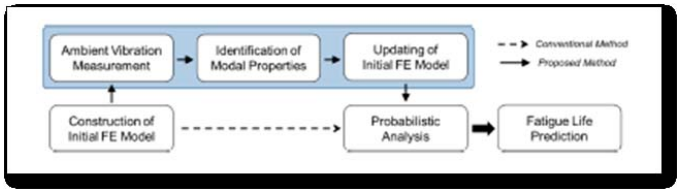


Figure 1. Schematic flow of procedure of the standard SHM

Figure 1 shows a schematic flow of procedure of the standard SHM. Conventionally, SHM procedure began at the construction of the initial FE Model then followed by probabilistic analysis of the data obtained then the determination of fatigue life prediction. This conventional method of damage detection consists of errors that can be avoided and produce more accurate results. On the other hand, the researchers proposed another method that could increase efficiency and locate damage accurately. This method consists of (1) the construction of an initial FE model of the desired bridge based on the provided plans. (2) Measure the ambient vibration of the bridge under normal vehicle traffic; (3) identify the modal properties which includes the natural frequencies, mode shapes and modal damping ratios form the obtained acceleration of data through the use of an output based modal identification method. (4) Updating the linear parameters of the initial FE model by using the identified modal properties. Lastly, (5) performing the probabilistic analysis of interest using the recently updated FE model.

III. BRIDGE MODEL

The bridge is a reinforced concrete bridge built in the 1960's and became operational in the year 1963. Its length is about 410 meters, linking Pandacan and Paco, Manila. In 2000, the Former President Joseph "Erap" Ejercito Estrada, changed the name of the Nagtahan – R. Magsaysay Link Road to Padre Jacinto Zamora Bridge under Proclamation No. 228.



Figure 2. Padre Jacinto Zamora Flyover (formerly known as Nagtahan – R. Magsaysay Link Road).

The existing concrete bridge considered in this study had existing hard copy of plans. The Bureau of Design of the Department of Public Works and Highways provided these plans.

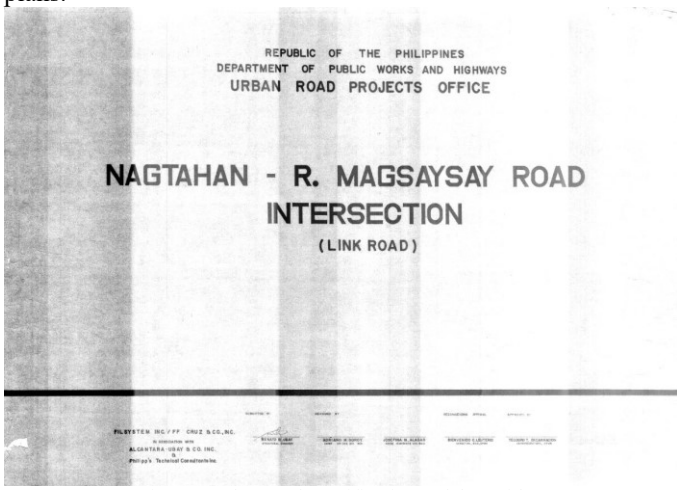


Figure 3. Plans and specifications of the Bridge

For the threshold limit determination, the bridge was modelled using a software called Staad Pro Vi8.

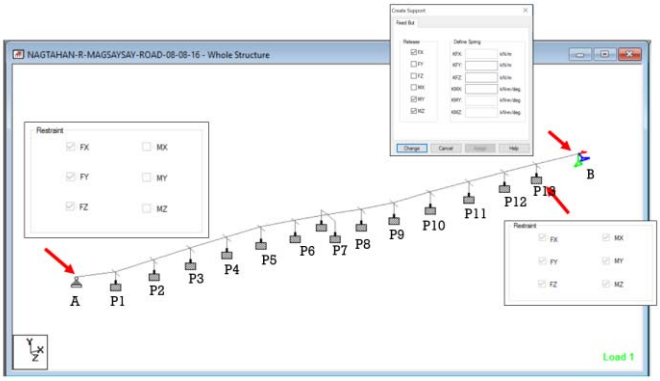


Figure 4. STAAD model

In Figure 4, Abutment A is modelled as a pinned support. The node is restrained in FX, FY, and FZ, which represents a pin support. On the other hand, Abutment B is modelled as a roller support. The releases are at FX, MY, and MZ which represents a roller support. In this case, since Abutment A is pinned support and Abutment B is roller support, the bridge is modeled as simply supported. In Figure 3.3., the columns are modelled as fixed support which means that the support is restrained at FX, FY, FZ, MX, MY, and MZ. Figures 3.4 and 3.5 below shows the elevation view and the plan view of the bridge model. Also note that the measurements and specifications were taken from the existing plans provided by DPWH Bureau of Design.

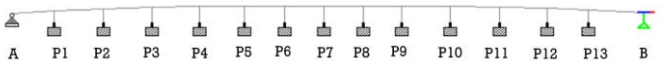


Figure 5. Elevation view

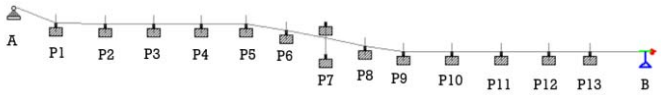


Figure 6. Plan view

Figure 7 below shows the specifications of the elements of the bridge. The piers are rectangular in cross section and the capping beam is a reverse T.

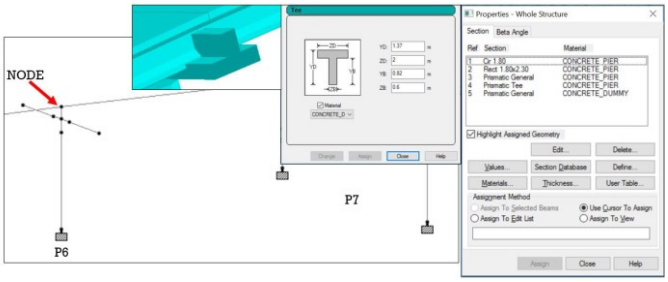


Figure 7. Properties of the elements of the bridge

Also, dummy elements were modelled as shown in Figure 8. The dummy elements have been set with properties shown below, to represent a very stiff element. This very stiff element would transmit the loadings from the superstructure down to the substructure.

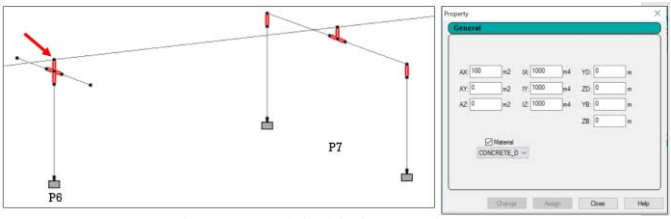


Figure 8. Model with dummy elements

The superstructure is modelled as a stick as shown in Figure 9. The stick model has specifications that were predetermined using the cross section of the four girders of the bridge including the deck. Since the deck and the girder have different compressive strength of concrete, the properties of the deck was transformed using the modular ratio, which then generated accurate section properties for the stick model of the superstructure.

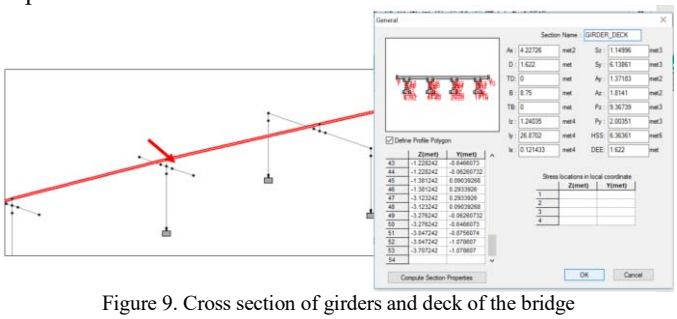


Figure 9. Cross section of girders and deck of the bridge

The application of loads are shown in Figure 10.

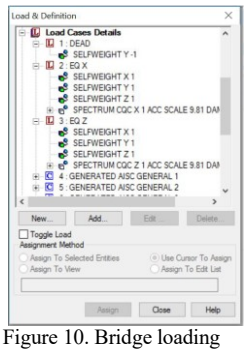


Figure 10. Bridge loading

IV. CALIBRATION OF SENSORS

A. *Objectives*

- To determine the scale factor of an accelerometer
- To have an accurate output reading from accelerometers without bias
- To get equal output reading between two different accelerometers tested under same conditions, after applying their corresponding scale factors.

B. *Procedure*

1. Gather output reading from the accelerometer for 500ms and determine the mean and the standard deviation.

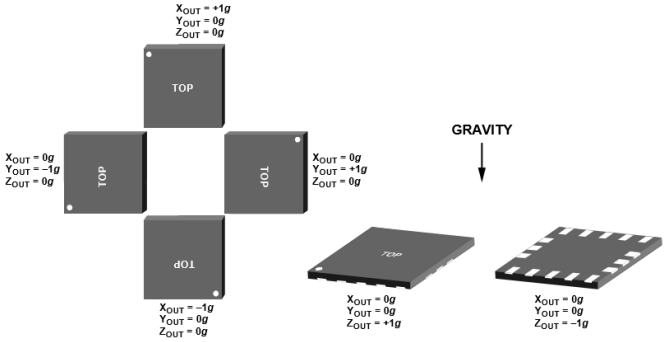


Figure 11. Reading of accelerometer for 500ms

At Orientation # 1: + Z - Axis Up

Axis	X-axis		Y-axis		Z-axis	
	ADXL 1	ADXL 2	ADXL 1	ADXL 2	ADXL 1	ADXL 2
Reading in LSB	9.96	6.88	17.92	27.75	230.5	240.17
Standard Deviation	0.6757	0.7253	0.8122	0.6614	0.9574	1.14

At Orientation # 2: - Z - Axis Up

Axis	X-axis		Y-axis		Z-axis	
	ADXL 1	ADXL 2	ADXL 1	ADXL 2	ADXL 1	ADXL 2
Reading in LSB	27.5	5.13	1	13.17	-281.25	-267.5
Standard Deviation	0.5774	0.6654	0.5	0.6236	0.7217	0.8660

At Orientation # 3: + X - Axis Up

Axis	X-axis		Y-axis		Z-axis	
	ADXL 1	ADXL 2	ADXL 1	ADXL 2	ADXL 1	ADXL 2
Reading in LSB	284.57	278.13	26.22	10.88	-32.09	-0.33
Standard Deviation	0.7704	0.6654	0.6565	0.5995	0.8295	0.6872

At Orientation # 4: - X - Axis Up

Axis	X-axis		Y-axis		Z-axis	
	ADXL 1	ADXL 2	ADXL 1	ADXL 2	ADXL 1	ADXL 2
Reading in LSB	-241.09	-245.26	-0.22	30.70	-42	-24.43
Standard Deviation	0.7753	0.6736	0.5866	0.6875	0.8847	0.9244

At Orientation # 5: + Y - Axis Up

Axis	X-axis		Y-axis		Z-axis	
	ADXL 1	ADXL 2	ADXL 1	ADXL 2	ADXL 1	ADXL 2
Reading in LSB	12.26	29.35	283.09	288.17	-24.65	-14.48
Standard Deviation	0.7353	0.6984	0.7171	0.6360	0.9141	0.8272

At Orientation # 6: - Y - Axis Up

Axis	X-axis		Y-axis		Z-axis	
	ADXL 1	ADXL 2	ADXL 1	ADXL 2	ADXL 1	ADXL 2
Reading in LSB	41.09	3.22	-249.30	-244.48	-22.78	-15.52
Standard Deviation	0.5833	0.5070	0.5465	0.5801	0.8318	0.9264

2. Determine the Scale Factor for each axis using the formula below:

Axis	Scale factor	
	ADXL 1	ADXL 2
X-axis	1.02678	1.02225
Y-axis	1.03982	1.04033
Z-axis	0.99951	0.99154

3. Determine the shift of each axis away from the typical value (shift up of reading at 1 gee and shift down of reading at -1 gee)

Axis	ADXL 1		ADXL 2	
	Shift up	Shift down	Shift up	Shift down
X – Axis	28.57	-14.91	22.13	-10.74
Y – Axis	27.09	-6.7	32.17	-11.52
Z - Axis	-25.5	25.25	-15.83	11.5

The aim on why the scale factor and amount of shift were recorded is for the reading of the accelerometer to be calibrated so that every accelerometer will have a unique output when tested under the same condition.

V. CALIBRATION OF SENSORS PART II

A. *Objectives*

- To determine the relationship between the acceleration grasp by the three axes of an accelerometer
- To create a mechanism to be integrated in the program that can compensate with tilt of orientation during installation

B. *Procedure*

Readings from an accelerometer were recorded at static condition for 5 seconds. At first try, the ADXL345 was situated in such a way that the z axis is perpendicular to the earth's surface while the x and y axis to be and in the second try, the x axis is no longer parallel to the earth's surface.

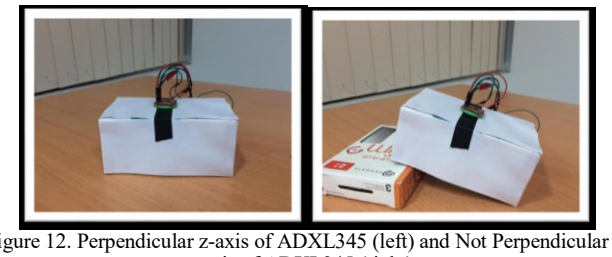


Figure 12. Perpendicular z-axis of ADXL345 (left) and Not Perpendicular z-axis of ADXL345 (right)

Mean	**18.67556**	**18.05778**	**241.6**	**71.12444**	**16.44**	**71.12444**
SD	**0.677628**	**0.654043**	**0.918937**	**0.648127**	**0.637495**	**0.648127**
Min	**18**	**17**	**240**	**70**	**15**	**70**
Max	**19**	**19**	**243**	**73**	**18**	**73**

Since the reading along the x and y axes in trial 1 is noticeable, and impossible to due to some random vibration since 19 LSB is approximately 0.07 gee. If we will relate this to the Mercalli scale for earthquake developed by the US Geological Survey as shown in the table below, 0.07 gee is approximately intensity 5 earthquake. This is very evident that the reading across the x and y axes are due to a misalignment of the accelerometer with respect to the earth's surface.

The aim here is to rotate the three axes in such a way that the z axis will land exactly on the location of the resultant vector. For this instance, the resultant vector is the gravitational acceleration.

Using the principle of rotation of axes, the following equations were derived and used to determine the rotated output acceleration in the three axes.

To date, the cost of sensor development per piece ranges from PHP 30,000.00 to PHP 100,000.00 depending on the caliber of accelerometers that will be used in the sensor assembly.

Mean	**0**	**0**	**243**	**0**	**0**	**247**
SD	**0.58**	**0.74**	**0.92**	**0.61**	**0.67**	**0.82**
Min	**-1**	**-1**	**241**	**-1**	**-1**	**246**
Max	**1**	**1**	**244**	**2**	**2**	**250**

VI. Results and Discussion

The research was done by initially modeling and analyzing the chosen flyover under static state. In doing proper static modeling using STAAD Pro V8i, the material properties of existing flyover were procured and shown. The enlistment includes all the items or materials used in the construction of the flyover. More of which, the concrete compressive strength and Young's Modulus of each element-material is defined and were secured prior to modeling. The dead loads for the analysis were from self-weight of the beams, diaphragms, parapets, premixes, concrete slab and other materials that will be given in details under the results and discussion of this paper. A general dimension for the length and width is also part of the modeling. As part of the secondary objectives, a step-by-step procedure on how the flyover was modeled under STAAD V8i Pro was written and made known. Parts of this include how a skeletal statically determinate structure should be modeled to follow STAAD V8i Pro design schematic.

In achieving true seismic response of the modeled bridge or hence syncing the theoretical data to the actual sensor reading also called as the independent variable of this paper, analysis using single degree of freedom (SDOF) of the flyover under free vibration without damping using STAAD V8i Pro was done upon running the model. The earthquake excitation and bridge responses is subjected to earthquake ground motion in the form of ground acceleration or Peak Ground Acceleration, (PGA). However, there's no guarantee that theoretical readings given by the software will automatically be in sync with the actual sensor readings. Calibration is done through the correction of sensor data. Translational acceleration load are used to simulate the ground motion of the time history analysis acceleration record in STAAD V8i Pro. STAAD B8i Pro software assumes complete fixity to all supports (since the modified prototype bridge pier model are constructed as monolithic structure). Then STAAD V8i Pro automatically computed the acceleration loads at each nodes and structural model. During analysis, acceleration couples with mass and STAAD.Pro has distributed the mass of the model to all joints. Each set of the single degree of freedom (SDOF) analysis, acceleration vs. time function will be plotted. The tabulated data of time (s) and acceleration amplitude (mm/sec2) is plotted using Microsoft Excel.

Acknowledgment

I like to express my deepest and sincerest gratitude to the following, who lend a hand in completing my research.

First, the Almighty God, for granting me his divine providence which helped me finish this paper. I would not be able to overcome all the struggles in making this research without the One who gave me strength all throughout the way.

Second, to DOST-PCIEERD and DPWH-BOD. Without their continuous support, I was able to materialize and perform my research. To them, thank you!

Third, to my thesis advisers Dr. Uy and Engr. Carreon who are co-authors of this paper. With their guidance and motivation, this paper was executed.

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